OverNet EDUCATION

18 NOVEMBRE

WPC® 20

ROAD TO 2021

DIGITAL EVENT

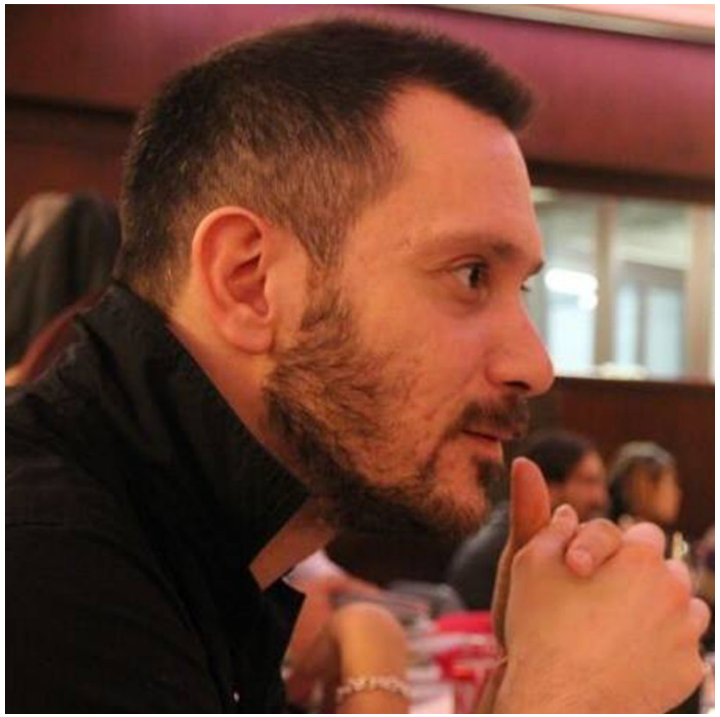
OverNet EDUCATION

Azure: quale engine SQL scegliere

Attenti a quei due

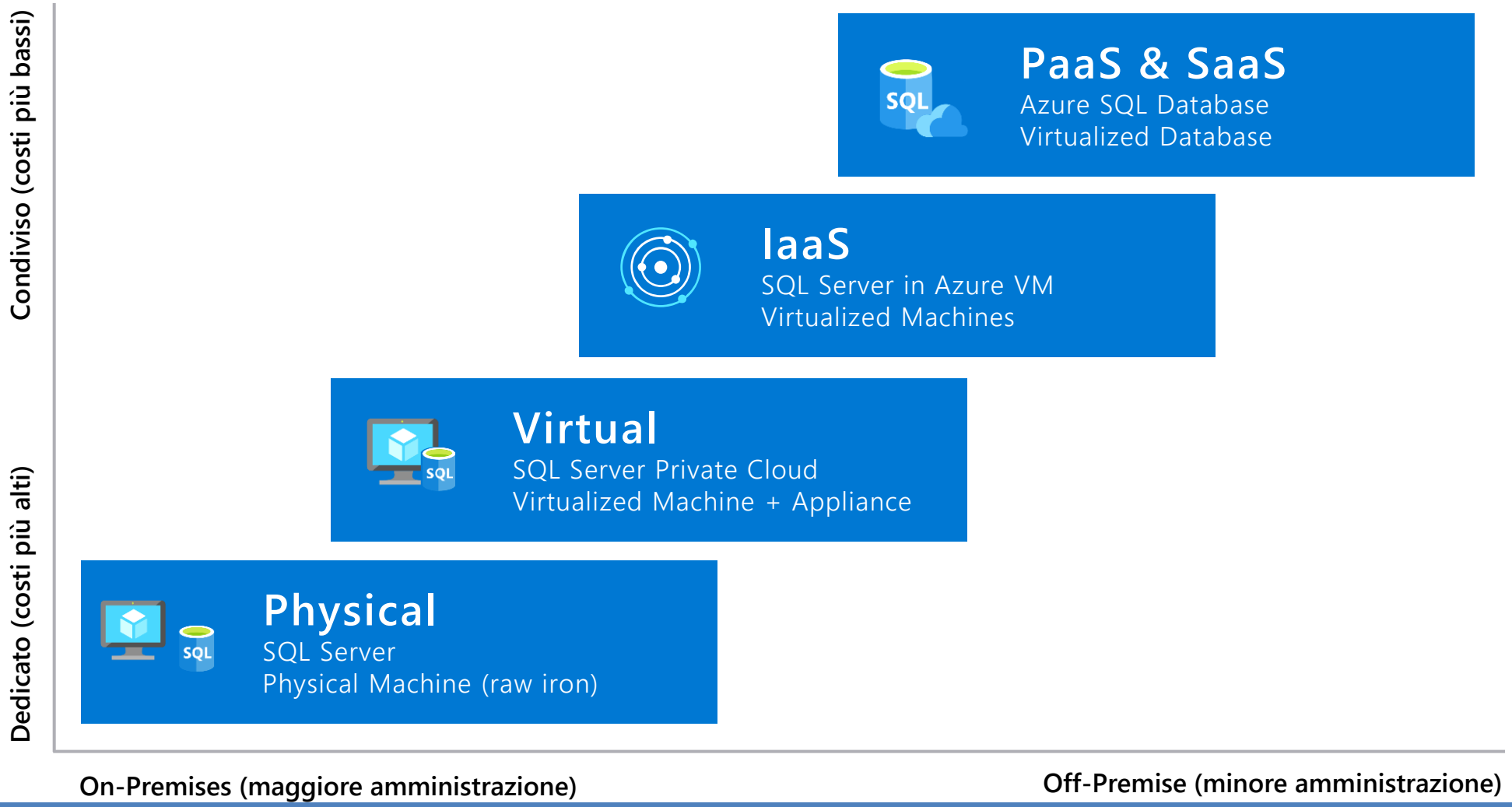
GIANLUCA HOTZ

LUCA FERRARI



Introduzione

Continuum piattaforma dati

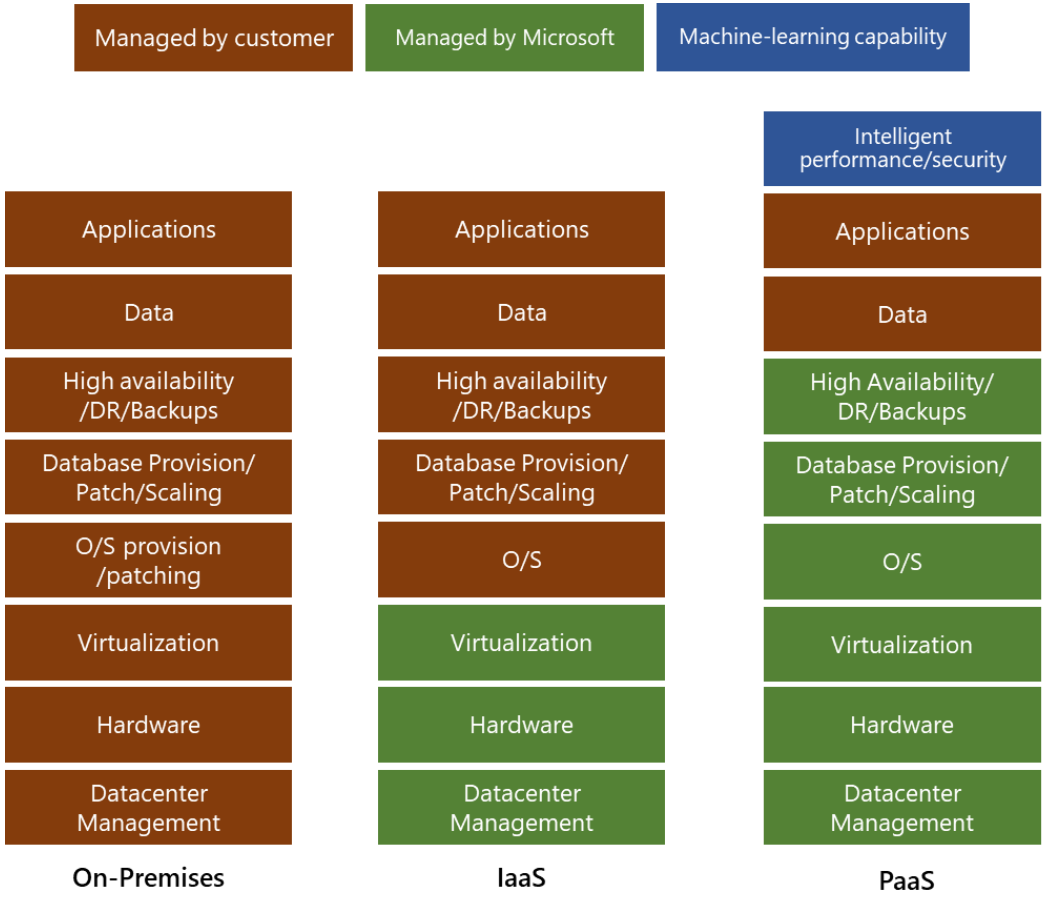


IaaS & PaaS

INFRASTRUCTURE AS A SERVICE

PLATFORM AS A SERVICE

Confronto servizi gestiti



Azure SQL Server Virtual Machines

Mantenere controllo Sistema Operativo e dimensionamento risorse

Disponibilità funzionalità che richiedono accesso al File System

- Es. Filestream, Semantic Search, Database Snapshot

Disponibilità altre funzionalità

- Es. Replication, Linked Server, Extended/OLE Stored Procedure, Policy Based Management

Disponibilità altri componenti SQL Server

- Es. SSIS, SSRS, SSAS, MDS, DQS, SQL Server Agent (c'è con Managed Instance)

«Extended Security Updates» gratuito

- Windows Server 2008/SQL Server 2008 R2 con supporto patch di sicurezza
- Rinvio modernizzazione (Luglio 2019 + 3 anni…)

Azure SQL Server VM Gallery

Versioni Sistema Operativo

- Windows, Ubuntu, Red Hat Enterprise Linux, SUSE Linux Enterprise Server

Versioni SQL Server

- Windows: 2008 R2 SP3, 2012 SP4, 2014 SP2, 2016 SP2, 2017, 2019
- Linux: 2017, 2019

Edizioni

- Enterprise, Standard, Web, Express, Developer

Fatturazione SQL Server

- «Pay-as-you-go» (a consumo insieme al resto)
- «Bring Your Own License» (a consumo solo S.O. e risorse VM)
- «Azure Hybrid Benefit»

Azure Virtual Machines News

Nuove immagini SQL Server nella Gallery

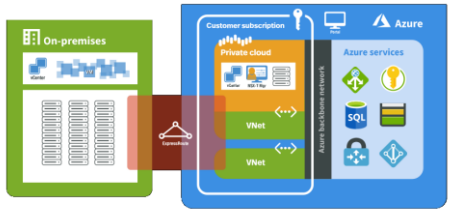
- SQL Server Reporting Services
  - Versioni 2016, 2017 e 2019, Edizioni Enterprise e Standard, con/senza Database Engine a bordo
- Analysis Service (Coming Soon!)

«Azure Dedicated Host»

- https://docs.microsoft.com/en-us/azure/azure-sql/virtual-machines/windows/dedicated-host

«Azure VMWare Solution»

- https://docs.microsoft.com/en-us/azure/azure-vmware



Gestione Azure SQL Server VM

SQL Server IaaS Agent extension

- https://docs.microsoft.com/en-us/azure/azure-sql/virtual-machines/windows/sql-server-iaas-agent-extension-automate-management
- «Automated Patching»
  - https://docs.microsoft.com/en-us/azure/azure-sql/virtual-machines/windows/automated-patching
- «Automated Backups»
  - https://docs.microsoft.com/en-us/azure/azure-sql/virtual-machines/windows/automated-backup
- Gestione tramite portale (configurazione SQL Server)
- Integrazione «Azure Key Vault» (TDE, CLE, Backup Cifrati)
- Cambio di versione/edizione flessibile
- Cambio di tipo licenza flessibile

Azure SQL Database

Database Fruibile come servizio (PaaS)

- Hardware, sistema operativo e software gestiti da Microsoft
- Backup automatico completamente gestito (full, differenziale, log)
- Alta disponibilità completamente gestita (modalità diverse in base a livello servizio)

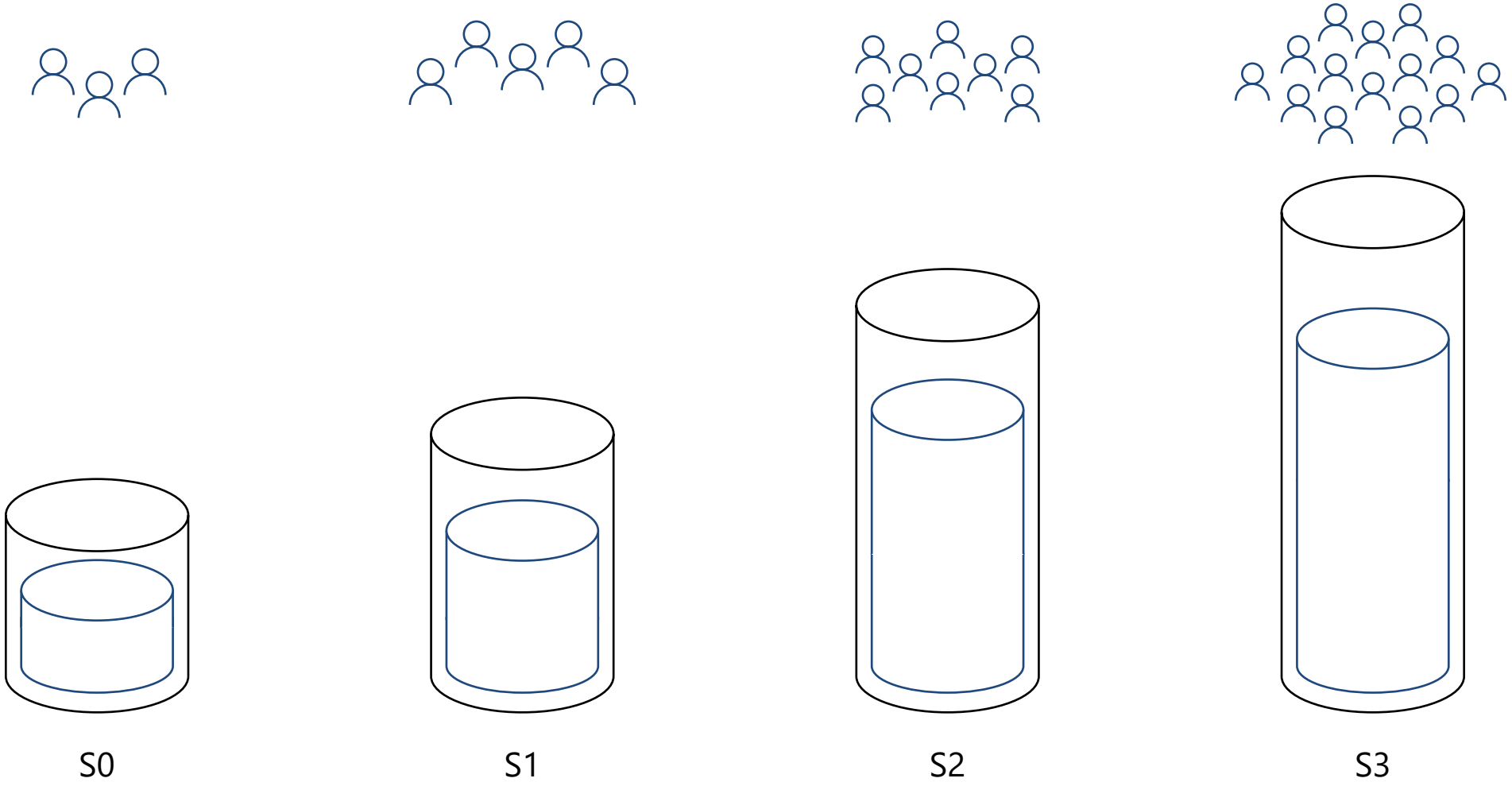
Dimensionamento per DTU o vCore

Diversi livelli di servizio

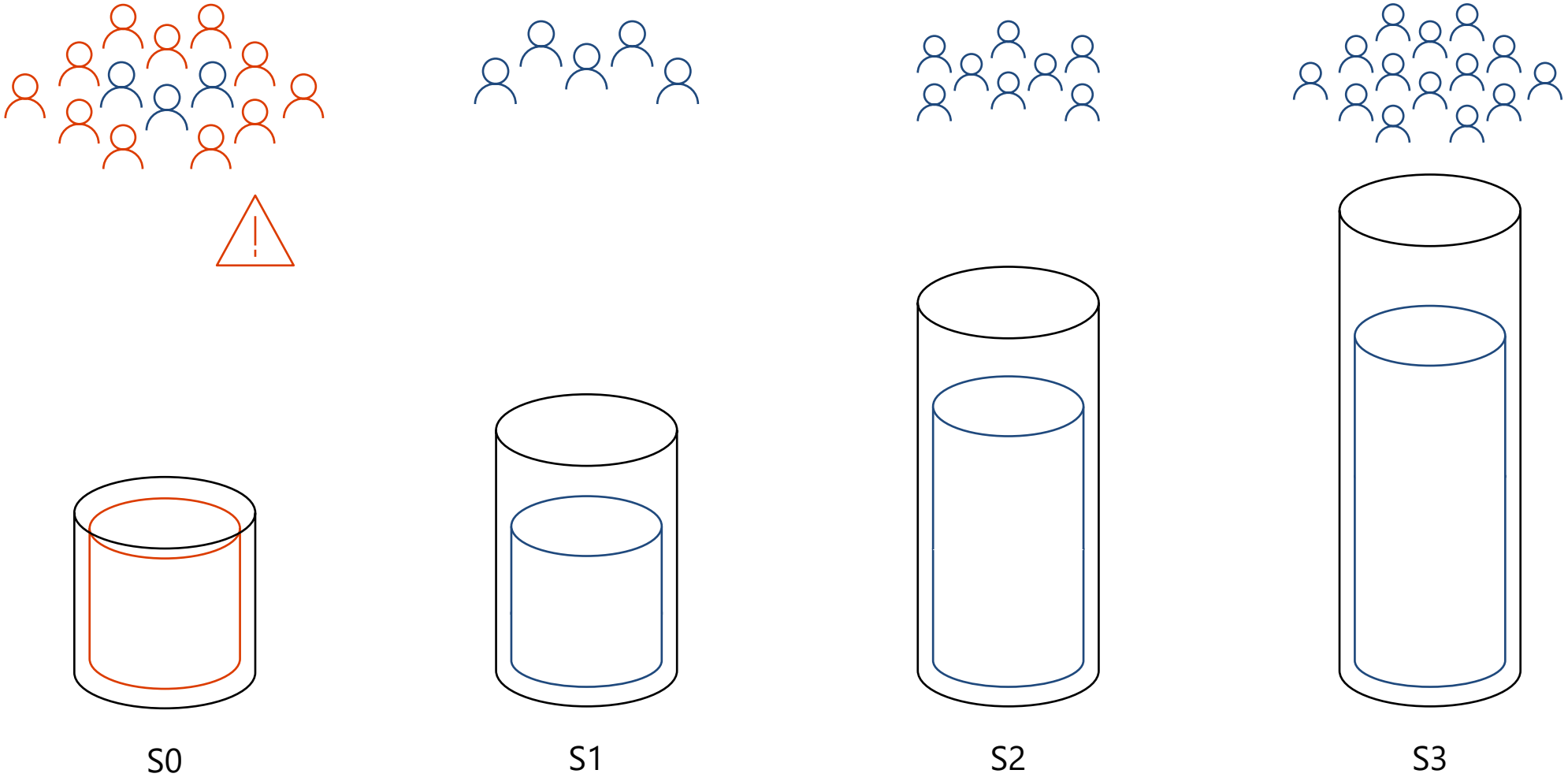
- Modello DTU: Basic, Standard, Premium
- Modello vCore: General Purpose, Business Critical

Scale Up/Down per database

Risorse preallocate e scalabilità



Risorse preallocate e scalabilità



«Elastic Database Pools»

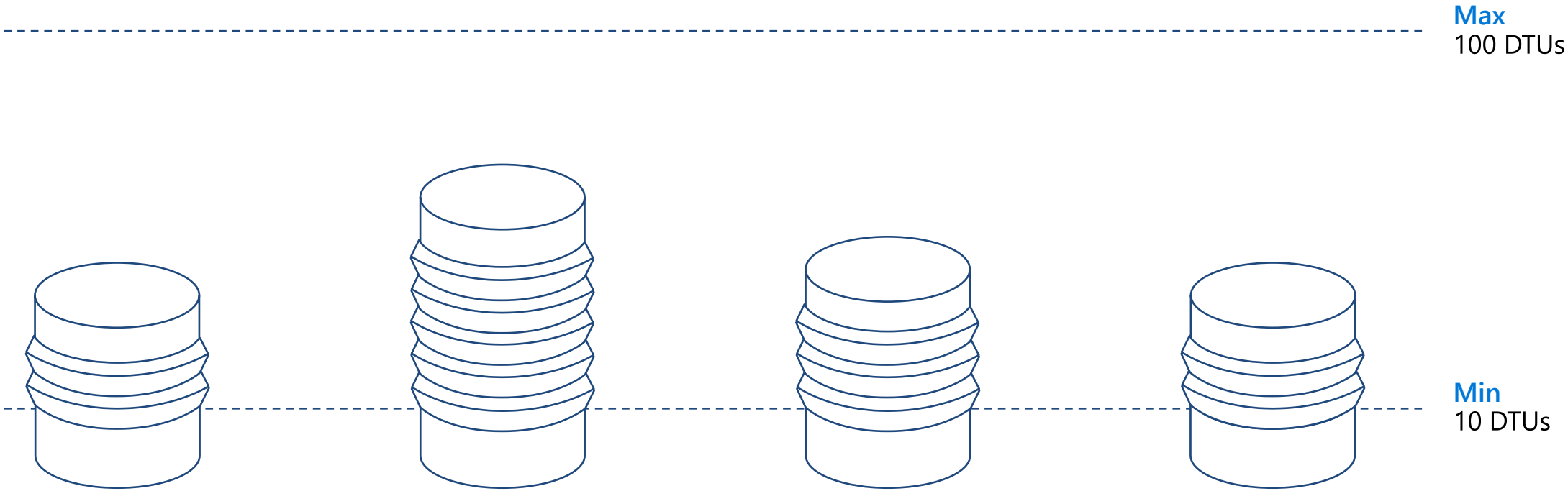
Pool di risorse condivise da più database

- DTU-Based
- vCore-Based

«Auto-Scale» database entro range definiti

Database aggiunti/rimossi a caldo

«Elastic Database Pools»



https://docs.microsoft.com/en-us/azure/sql-database/sql-database-elastic-pool

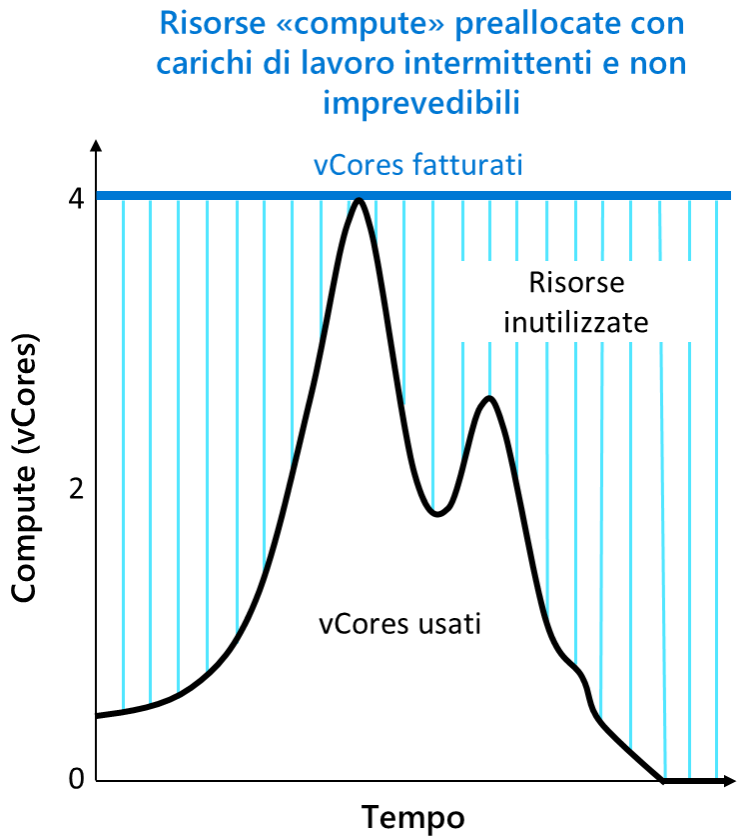
Inefficienza preallocazione CPU

Preallocazione

- efficace con pattern prevedibili

Inefficienze

- Sovrallocazione in periodi di utilizzo minore
- Sottoallocazione per ridurre costi
- Tempo speso a gestirla



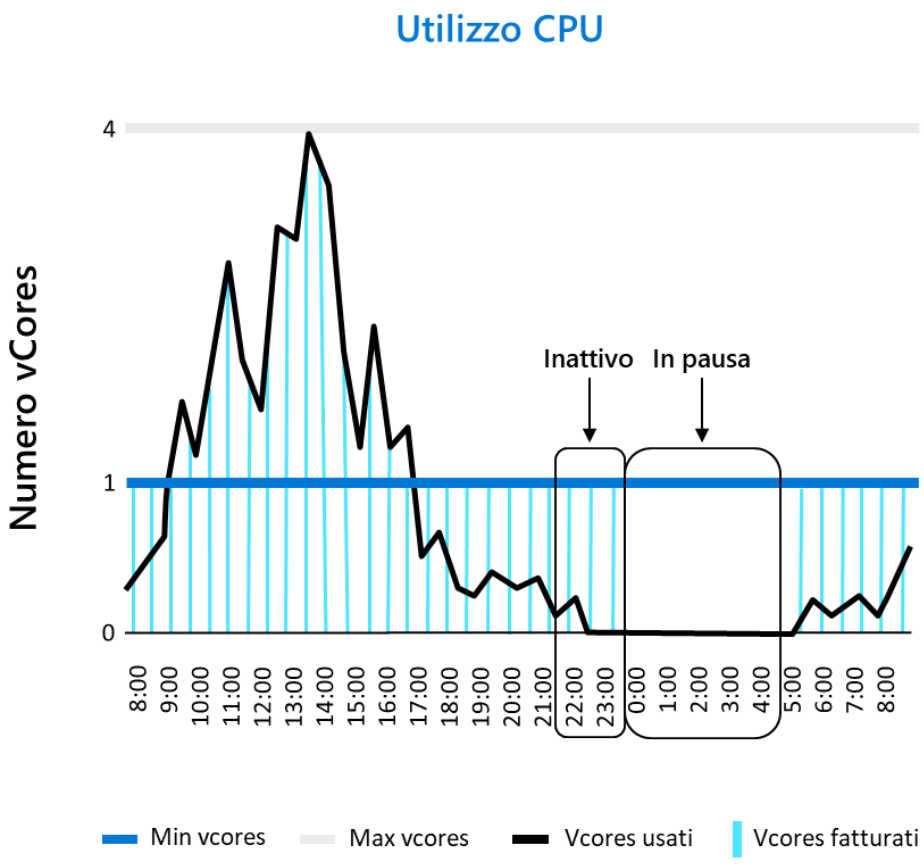
«Serverless»

Min/Max vCore (da 0,5 a 40)

«Autopause Delay»
- Costo zero
- Tempo di «warmup»

Fatturazione al secondo

Al momento solo
- General Purpose vCore
- Gen 5 Hardware



https://docs.microsoft.com/en-us/azure/sql-database/sql-database-serverless

Azure SQL Database Hyperscale

Nuovo «Service Tier» per il modello vCore-based

Supporta 100TB (di più… 100 sono quelli testati)

Prestazioni elevate, nuova architettura database engine

«File Snapshots»

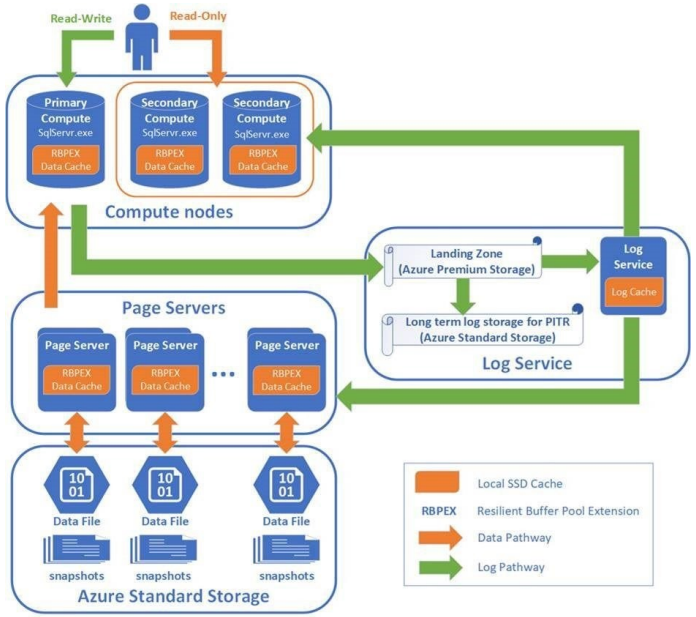
- Backup più veloci, hanno zero impatto sulle prestazioni
- Restore estremamente veloci

«Scale out» rapido

- Creazione di replica in sola lettura per redistribuzione carico di lavoro

«Scale up/down» rapido

- Tempo costante, minuti, sia «storage» che «compute»



Azure SQL Database Managed Instance

Opzione di Deployment di SQL Database

- Rappresenta ed espone l'intera istanza di SQL Server
- Molto differente «Azure SQL Database Logical Server»

100% copertura funzionalità «On-Premise» (quasi..)

- SQL Server Agent, Backup/Restore, Replication, CDC, CLR, Broker ecc.
  - https://docs.microsoft.com/en-us/azure/sql-database/sql-database-features

Mantiene tutti i vantaggi di un PaaS

- Completamente gestita
- «Scale up/down» semplice e rapido
- HA/DR già inclusi

Opzioni deployment Azure SQL Database

Azure SQL Database

Single

Modello a singolo database con prestazioni carico di lavoro prevedibili

Best for
applicazioni che richiedono risorse garantite a livello di database

Elastic Pool

Modello molteplici database con risorse condivise, ottimizzato per maggiore efficienza applicazioni multi-tenant

Best for
Applicazioni in modalità SaaS con molteplici database che possono condividere risorse

Managed Instance

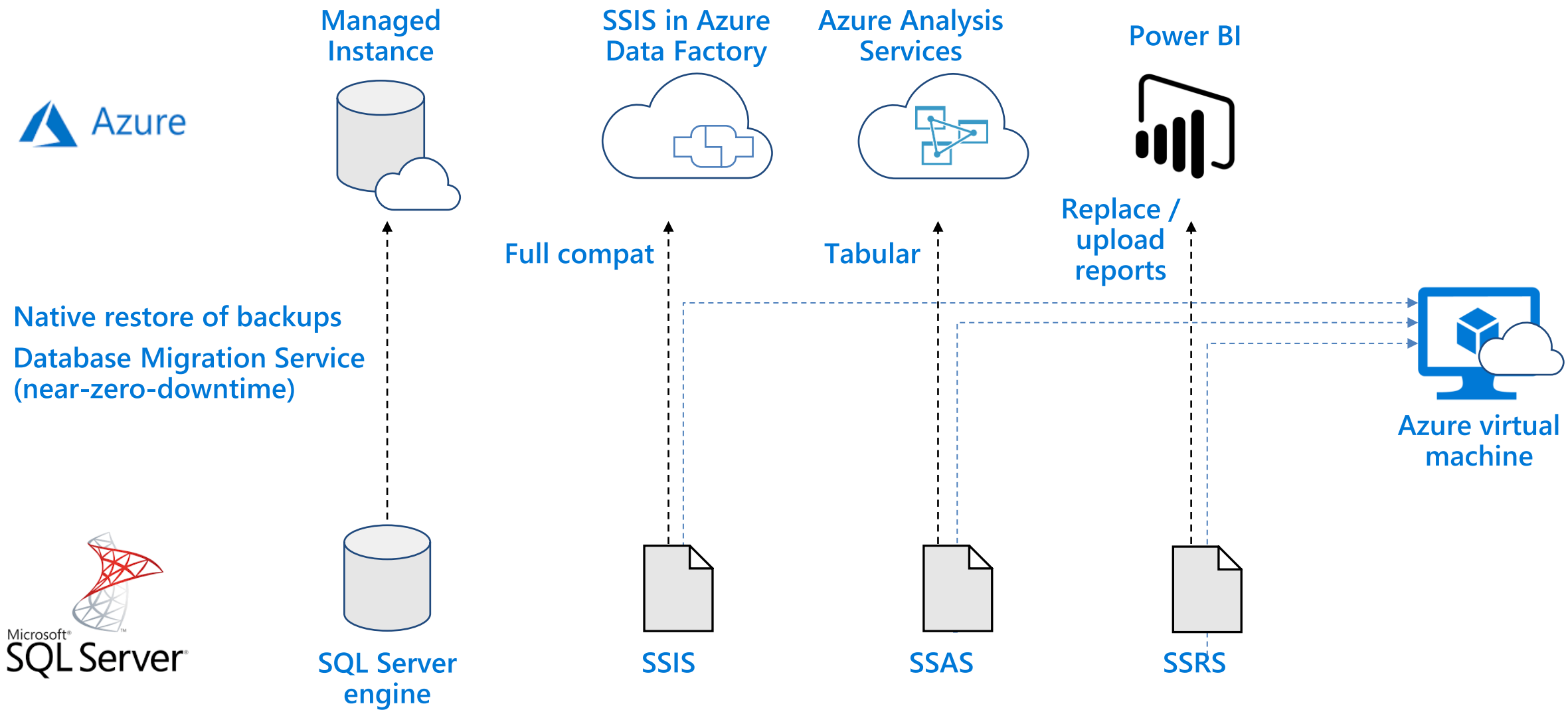
Modello a istanza con elevata compatibilità con SQL Server IaaS mantenendo i vantaggi di un PaaS

Best for
modernizzazione su larga scala con minori sforzi e attriti

Service Tiers

- General Purpose
- Business Critical
- Hyperscale
- Serverless

Migrazione On-Premise a IaaS/PaaS



- Native restore of backups
- Database Migration Service (near-zero-downtime)

Azure OSS SQL Managed Database

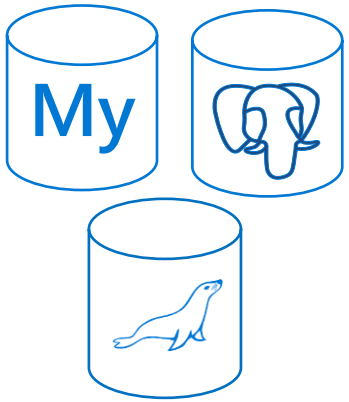
Azure Database for PostgreSQL

- Single Server
- **Flexible Server (Preview)**
- Hyperscale (Citus, sharding orizzontale)

Azure Database for MySQL

- Single Server
- **Flexible Server (Preview)**

Azure Database for MariaDB



Other Azure SQL Options

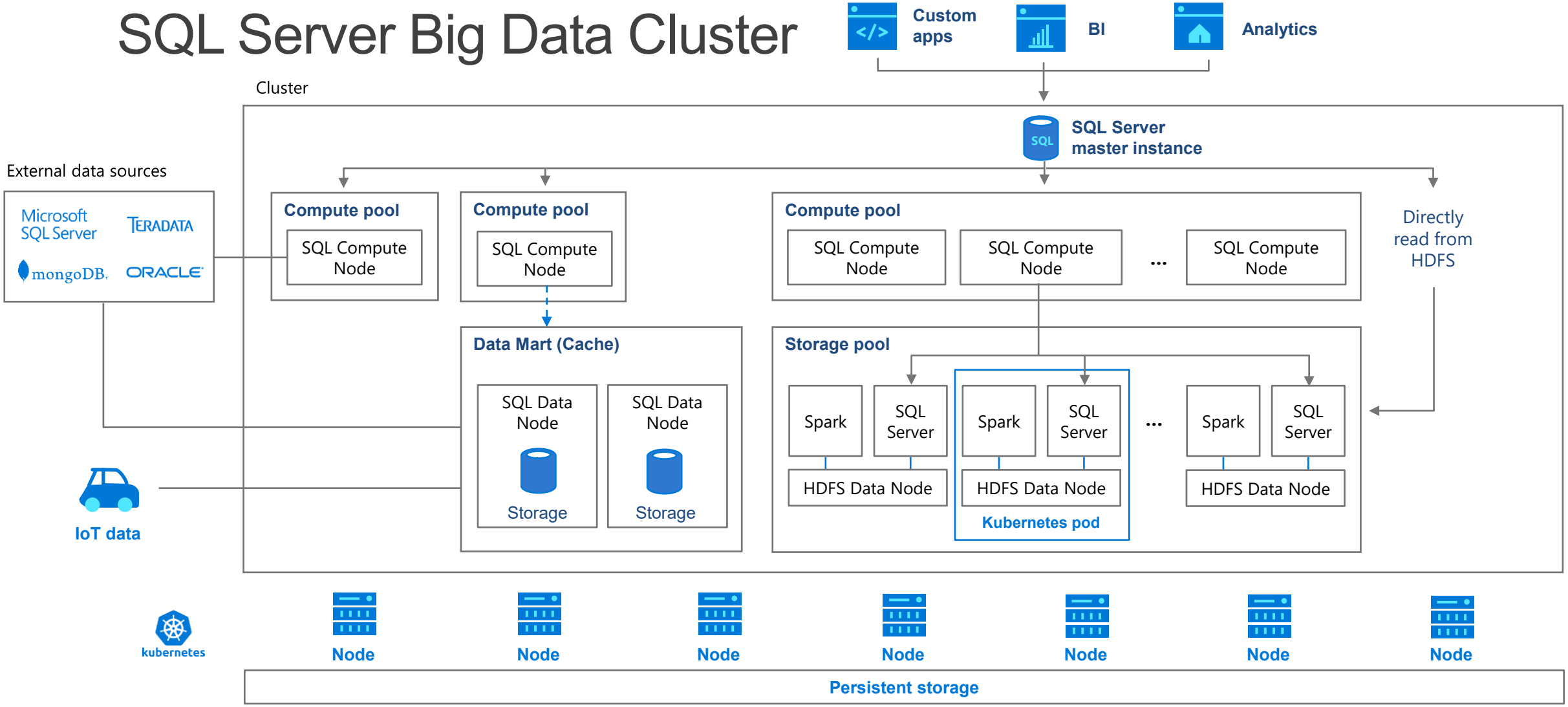
Azure Arc Enable Data Services

- SQL Managed Instance (Preview)
- Azure PostgreSQL Hyperscale (Preview)

Azure SQL Edge

Container in Azure Kubernetes Services

SQL Server Big Data Cluster



https://docs.microsoft.com/en-us/sql/big-data-cluster/big-data-cluster-overview

Synapse Analytics

Azure Synapse Analytics is a limitless analytics service, that brings together enterprise data warehousing and Big Data analytics. It gives you the freedom to query data on your terms, using either serverless on-demand or provisioned resources, at scale. Azure Synapse brings these two worlds together with a unified experience to ingest, prepare, manage, and serve data for immediate business intelligence and machine learning needs.

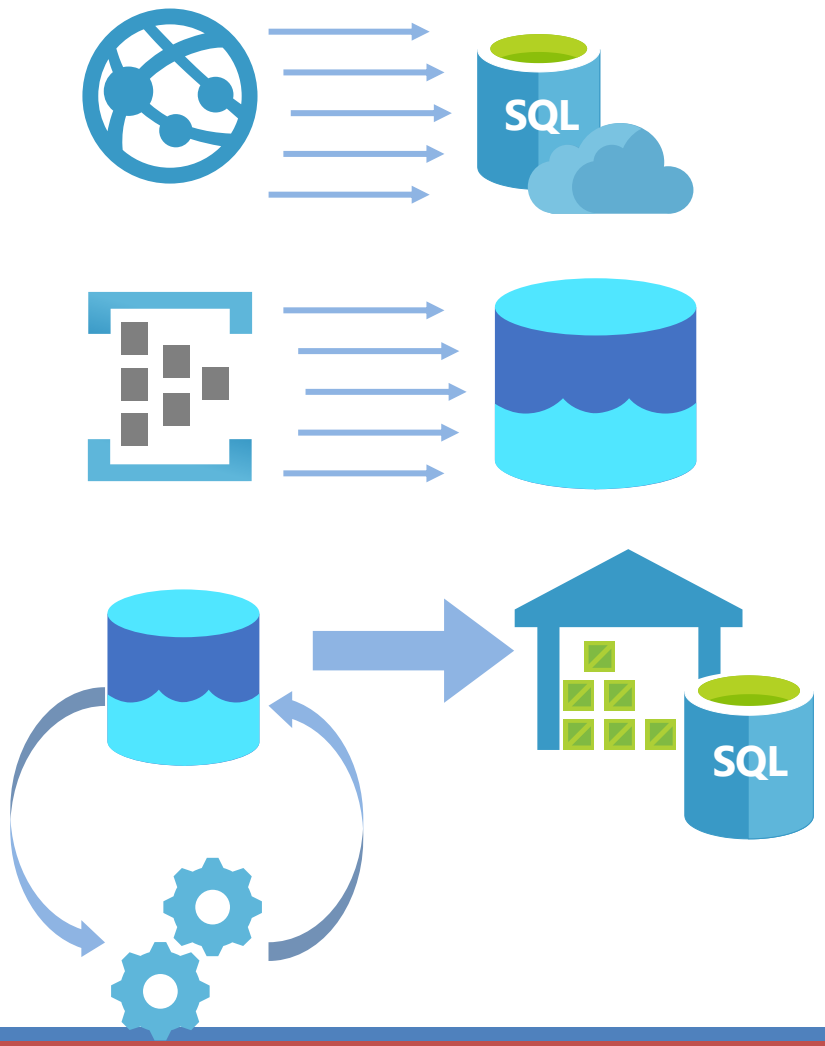
What workloads are NOT suitable?

Operational workloads (OLTP)

- High frequency reads and writes.
- Large numbers of singleton selects.
- High volumes of single row inserts.

Data Preparations

- Row by row processing needs.
- Incompatible formats (XML).



What Workloads are Suitable?

OverNet EDUCATION

Analytics

Store large volumes of data.

Consolidate disparate data into a single location.

Shape, model, transform and aggregate data.

Batch/Micro-batch loads.

Perform query analysis across large datasets.

Ad-hoc reporting across large data volumes.

All using simple SQL constructs.

Parallelism

OverNet EDUCATION

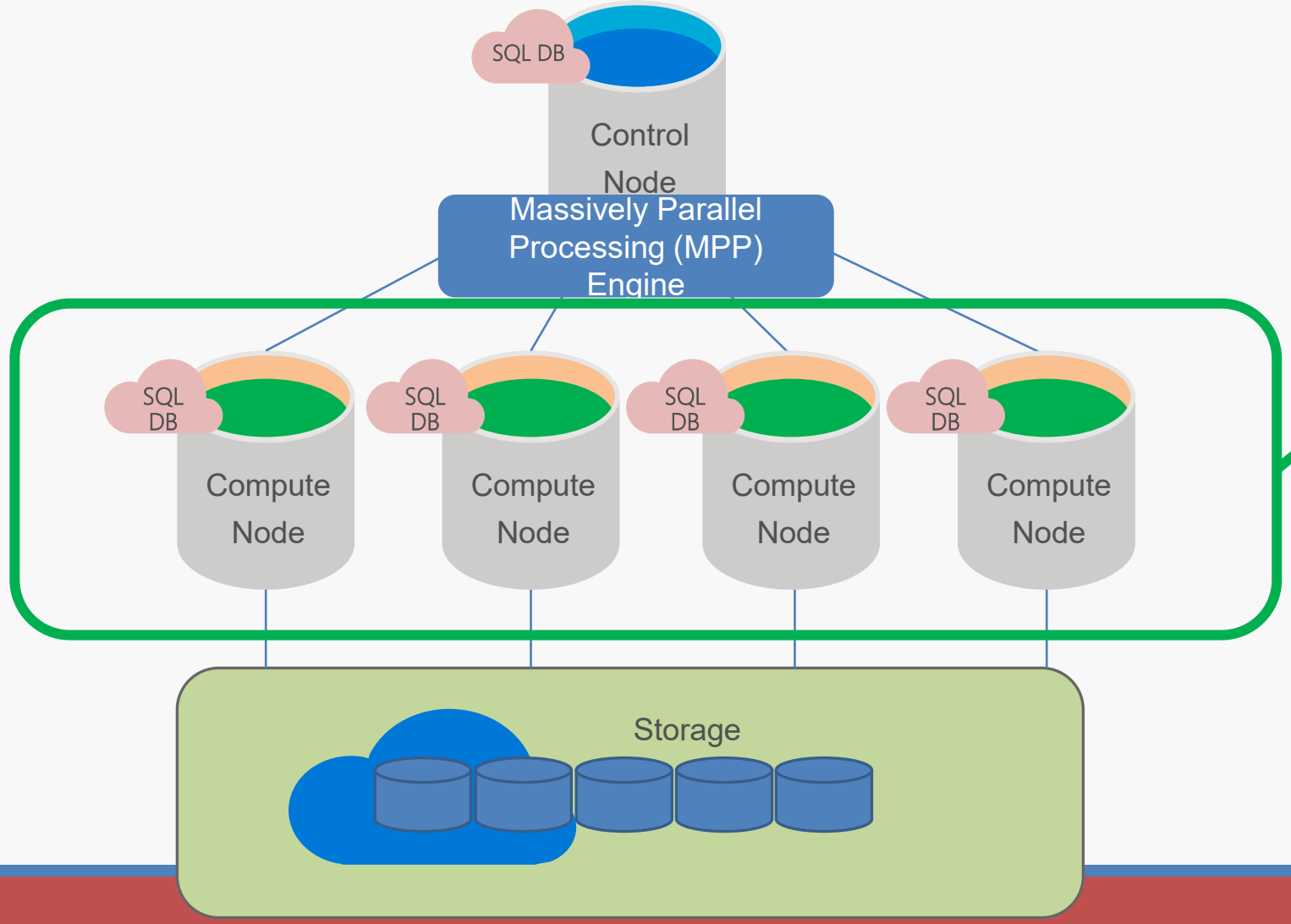
SMP - Symmetric Multiprocessing

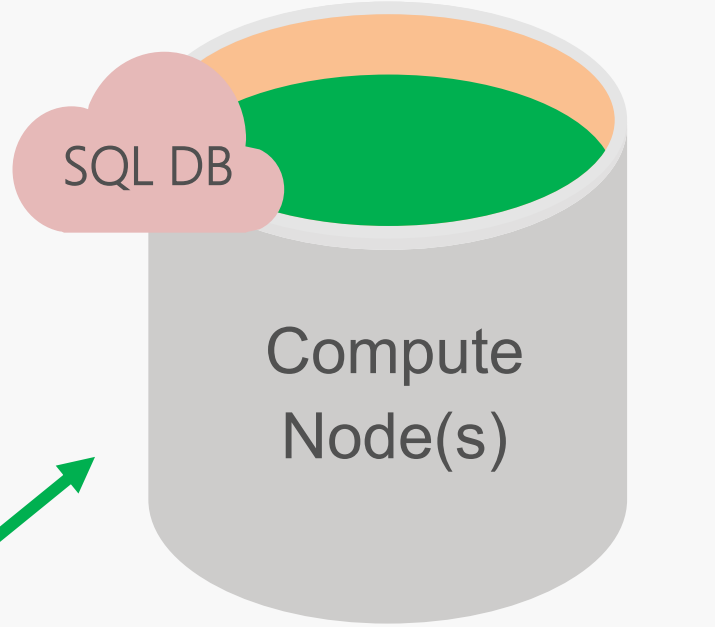
- Multiple CPUs used to complete individual processes simultaneously
- All CPUs share the same memory, disks, and network controllers (scale-up)
- All SQL Server implementations up until now have been SMP
- Mostly, the solution is housed on a shared SAN

MPP - Massively Parallel Processing

- Uses many separate CPUs running in parallel to execute a single program
- Shared Nothing: Each CPU has its own memory and disk (scale-out)
- Segments communicate using high-speed network between nodes

Azure Synapse Analytics (Sql Dw) - Compute Nodes

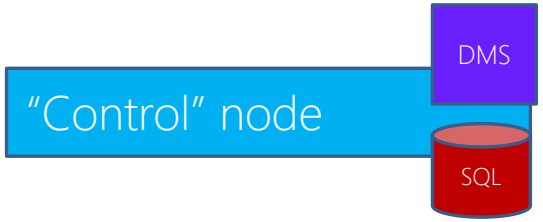




Azure SQL Database

An increase of DWU will increase the number of compute nodes

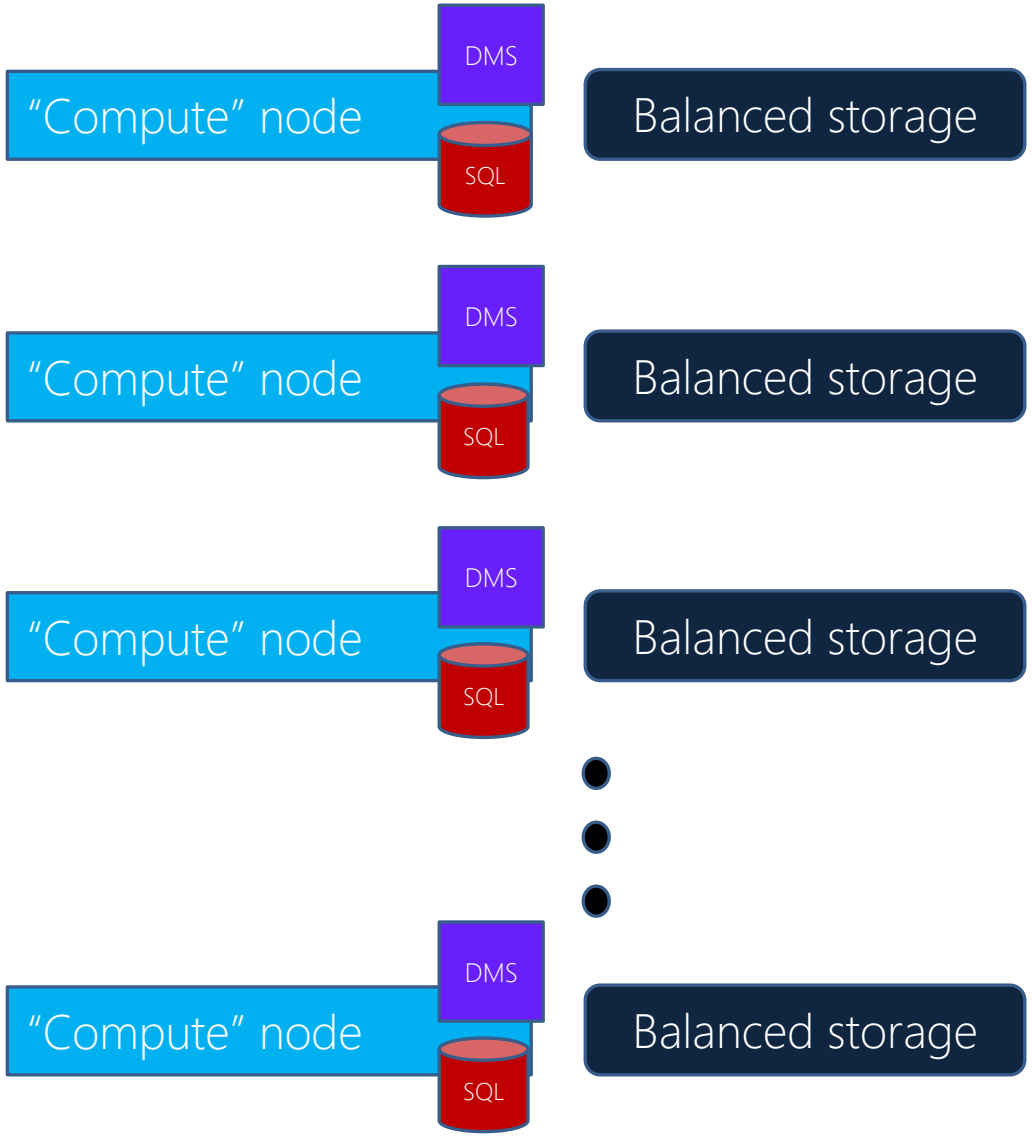
SQL DW Logical Architecture (overview)



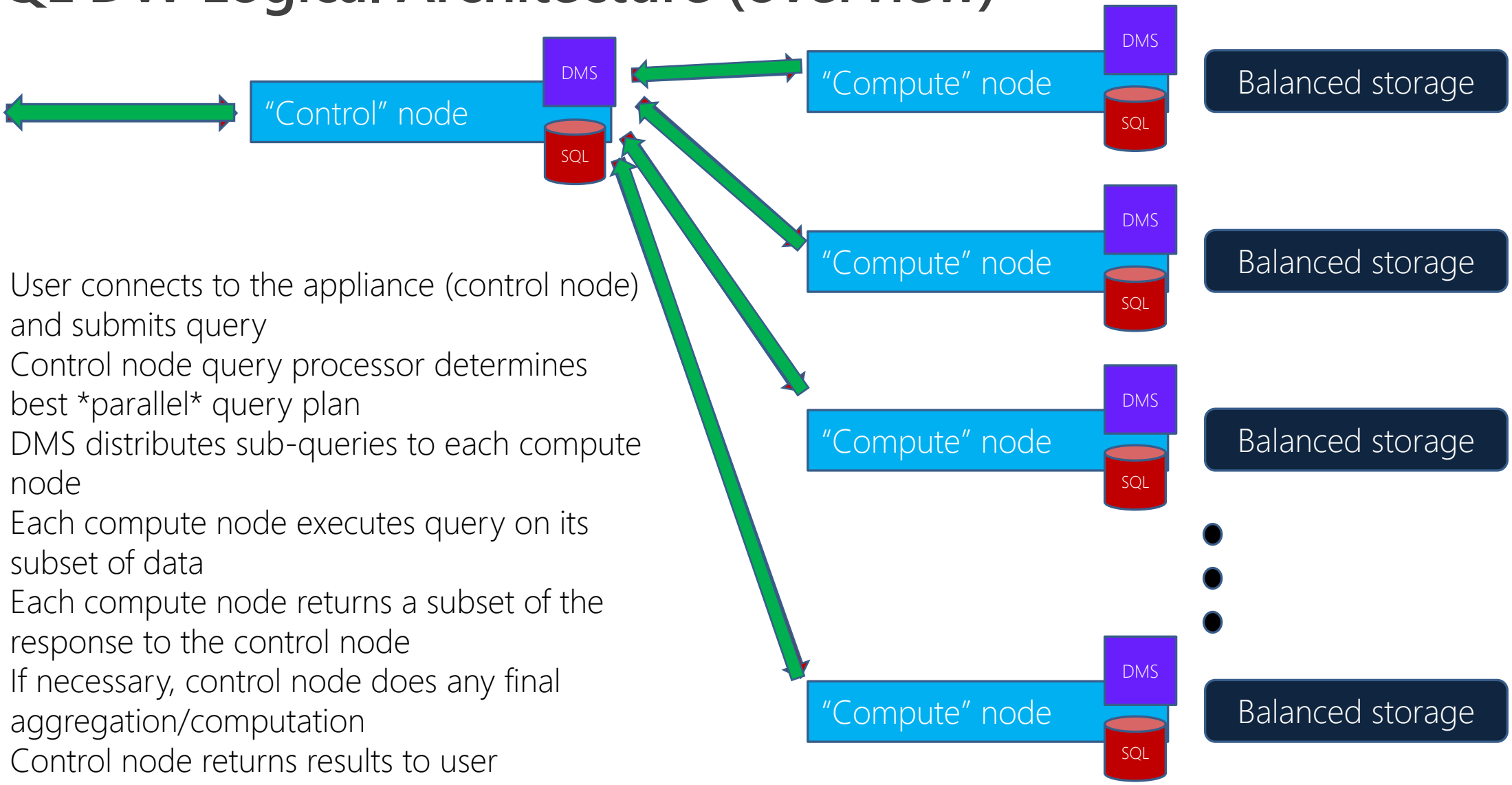
Compute Node – the "worker bee" of SQL DW
- Runs Azure SQL Server DB
- Contains a "slice" of each database
- CPU is saturated by storage

Control Node – the "brains" of the SQL DW
- Also runs Azure SQL Server DB
- Holds a "shell" copy of each database
  - Metadata, statistics, etc
- The "public face" of the appliance

Data Movement Services (DMS)
- Part of the "secret sauce" of SQL DW
- Moves data around as needed
- Enables parallel operations among the compute nodes (queries, loads, etc)



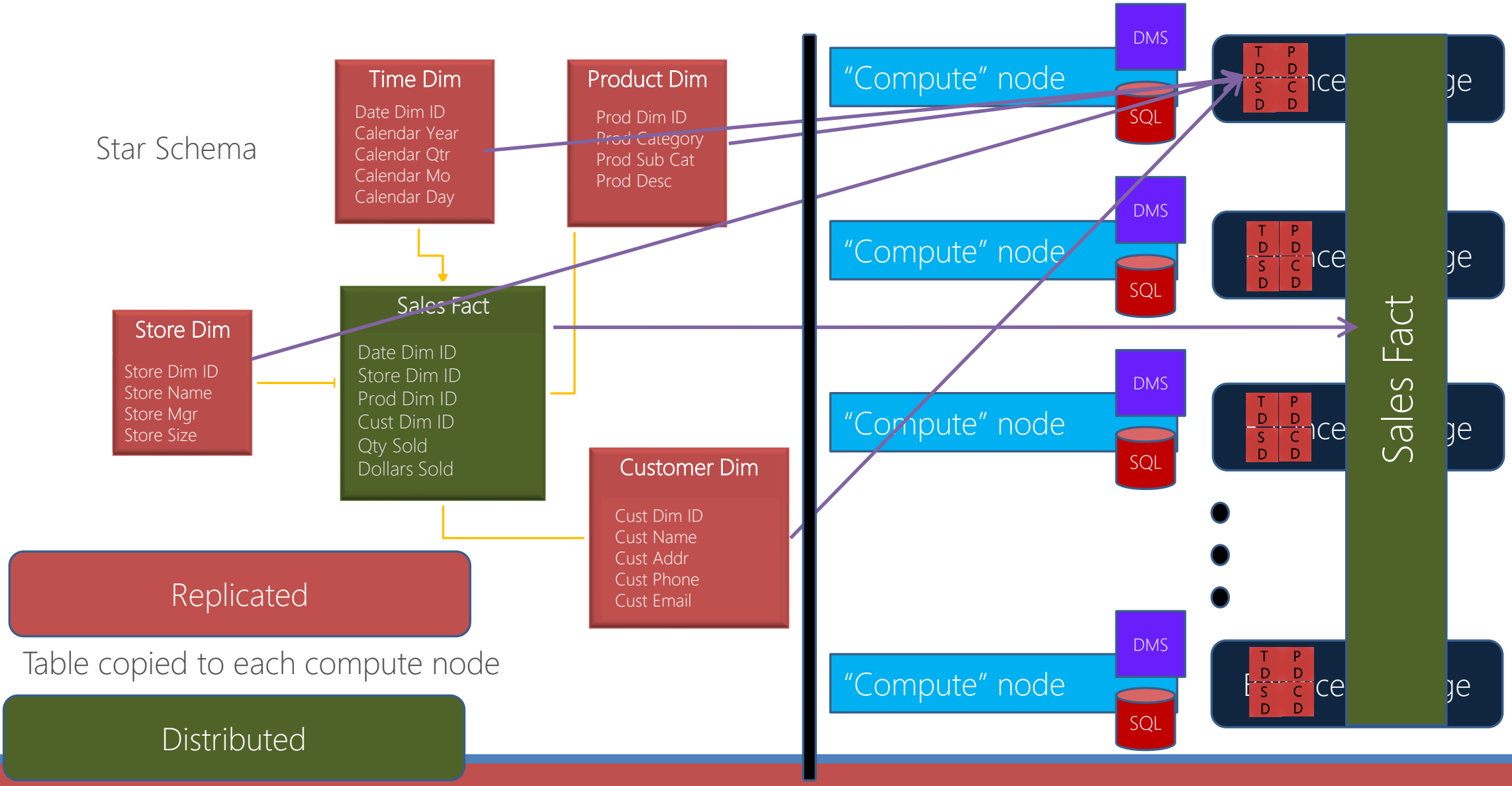
SQL DW Logical Architecture (overview)



1) User connects to the appliance (control node) and submits query
2) Control node query processor determines best *parallel* query plan
3) DMS distributes sub-queries to each compute node
4) Each compute node executes query on its subset of data
5) Each compute node returns a subset of the response to the control node
6) If necessary, control node does any final aggregation/computation
7) Control node returns results to user

Queries running in parallel on a subset of the data, using separate pipes effectively making the pipe larger

SQL DW Data Layout Options



Replicated

Table copied to each compute node

Distributed

Table spread across compute nodes based on "hash"

Tables – Distributions

OverNet EDUCATION

Round-robin distributed

Distributes table rows evenly across all distributions at random.

Hash distributed

Distributes table rows across the Compute nodes by using a deterministic hash function to assign each row to one distribution.

Replicated

Full copy of table accessible on each Compute node.

```
CREATE TABLE dbo.OrderTable
(
    OrderId   INT NOT NULL,
    Date      DATE NOT NULL,
    Name      VARCHAR(2),
    Country   VARCHAR(2)
)
WITH
(
  CLUSTERED COLUMNSTORE INDEX,
  DISTRIBUTION = HASH([OrderId]) |
                        ROUND ROBIN |
                        REPLICATED
);
```

Distributed Data = Buckets of Water

Evenly spreading the data leads to even use application resources

Common table distribution methods

Table Category	Recommended Distribution Option
Fact	Use hash-distribution with clustered columnstore index. Performance improves because hashing enables the platform to localize certain operations within the node itself during query execution. Operations that benefit: COUNT(DISTINCT(<hashed_key>)) OVER PARTITION BY <hashed_key> most JOIN <table_name> ON <hashed_key> GROUP BY <hashed_key>
Dimension	Use replicated for smaller tables. If tables are too large to store on each Compute node, use hash-distributed.
Staging	Use round-robin for the staging table. The load with CTAS is faster. Once the data is in the staging table, use INSERT…SELECT to move the data to production tables.

Where to start

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/massively-parallel-processing-mpp-architecture

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/cheat-sheet

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/create-data-warehouse-portal

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/design-elt-data-loading

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/what-is-a-data-warehouse-unit-dwu-cdwu

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-workload-management

https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/sql-data-warehouse-overview-manage-security

Polybase: https://docs.microsoft.com/en-us/sql/relational-databases/polybase/polybase-guide?view=sql-server-ver15

Contatti

OverNet Education
+39 02 365738
info@overneteducation.it
www.overneteducation.it
www.wpc2020.it

#wpc20it

ROZZANO - MILANO
BOLOGNA
ROMA
GENOVA
TORINO